

# Fashion Retrieval

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**Abstract**—The Composed Image Retrieval (CIR) task involves retrieving target images based on a composed query of a reference image and modified text. This task is challenging due to the scarcity of positive examples and the limited scope of negative sampling techniques. This study aims to improve the effectiveness of contrastive learning for CIR by introducing a multi-modal large language model to generate high-quality triplets and a two-stage fine-tuning framework. We leverage pretrained vision-language models, such as BLIP-2, to generate and concatenate sentence-level prompts with the relative caption. Our method scales the number of negatives using static representations. Our approach achieves state-of-the-art performance on the FashionIQ and CIR datasets, boosting the R@10 and R@50 metric across various baseline models. Additionally, our method enables robust zero-shot CIR capabilities. These advancements demonstrate the potential of scalable triplet construction and enhanced fine-tuning to drive significant progress in CIR tasks, even in low-resource scenarios. Injecting knowledge about angle transformations into the CIR model to distinguish between different view angles and perspectives holds significant potential.

**Index Terms**—composed image retrieval, contrastive learning, large language model, fine-tuning, zero-shot learning, FashionIQ, CIR

## I. INTRODUCTION

## II. RELATED WORK

**Composed image retrieval (CIR)** combines a query image with textual descriptions to retrieve target images, a challenging task requiring effective fusion of visual and textual modalities. Early approaches, such as Vo et al., 2019 [1], explored residual connection-based methods for integrating queries. Recent work leverages vision-language models: Liu et al., 2021 [2] introduced CIRPLANT with transformer-based adaptation, while Baldrati et al., 2022, 2023 [3], [4] utilized CLIP for feature fusion. Xu et al. [5] proposed Comquery-Former with a unified transformer architecture and global-local alignment. Zhao et al., 2022 [6] introduced progressive learning with adaptive weighting for hybrid queries, and Bai et al., 2023 [7] enhanced retrieval using sentence-level prompts from pretrained models. Other innovations include zero-shot methods (Saito et al. [8]), context-aware mapping (Tang et al. [9]), and application to video retrieval (Ventura et al. [10]).

Recent advances in **data generation for CIR** focus on synthetic data and multimodal retrieval techniques. CompoDiff [11], using latent diffusion, achieves state-of-the-art performance on benchmarks like Fashion IQ [12], and CIR [2],

through automated triplet generation. MagicLens [13] improves retrieval via self-supervised learning with synthesized embeddings, while InstructPix2Pix [14] generates text and images for compositional tasks using GPT-3 and diffusion models. Efforts like UniIR [15] integrate vision-language queries to enhance retrieval efficiency and accuracy across diverse datasets.

**Negative sampling** plays a vital role in contrastive learning for image retrieval. Feng et al. [16] used multi-modal language models to generate triplets for CIR, addressing positive data scarcity. Zhou and Li [17] proposed a coarse-to-fine alignment framework for cross-modal image retrieval, improving performance through targeted sampling. Contrastive learning approaches, such as SimCLR [18] and non-parametric instance discrimination [19], have shown that augmentations and effective sampling of negatives are essential for learning robust representations. Additionally, conditional negative sampling [20] enhances feature transferability to new distributions, while contrastive hashing with vision transformer [21] improves retrieval performance by integrating hard negative samples.

## III. METHODOLOGY

### A. Baselines

The baselines we used are derived from two key papers. Bai et al., 2023 [7] introduces the initial methodology for composed image retrieval (CIR) using sentence-level prompts. Feng et al., 2024 [16] presents improvements over the first by leveraging contrastive learning with scaling positives and negatives to enhance CIR performance. Figure 1, based on Figure provides an overview of the methodology.

1) *Preliminary*: Suppose a CIR dataset consists of  $N$  annotated triplets consisting of reference images ( $r$ ), modified text ( $t$ ), and target image ( $t$ ). A candidate image set consists of all reference images and target images. The task is to use reference image  $r(i)$  and modified text  $t(i)$  to compose a query  $q(i)$ . Then this  $q(i)$  is used to retrieve target images from the candidate set.

2) *Paradigm of Composed Image Retrieval*: Multiple annotated triplets are combined into a mini-batch, and the reference images and modified texts in the same batch are then encoded using a query encoder to obtain query representations. The target images are encoded using an image encoder to obtain target image representations. The cosine similarity is then

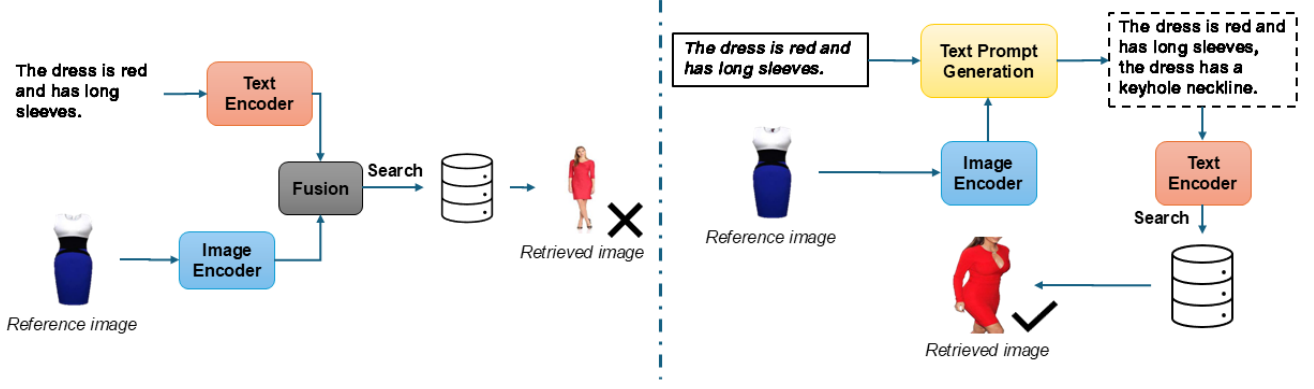


Fig. 1. Workflows of the Late Fusion method (left) and our proposed one (right).

adopted to calculate the similarity between the query and target image representations.

3) *Contrastive Learning*: Treat the annotated examples as positive examples and treat the examples obtained by replacing the target image in the positive examples with another image in the mini-batch as the negatives. This approach pulls the query and target representations in positive examples closer while pushing the query and target representations in negative samples further. This yielded good results; however, the lack of positive examples and negative examples severely limits the full performance of contrastive learning, leading to optimization.

4) *Scaling Positive Examples*: Using multi-modal Large Language Model (MLLM) to construct the triplets for CIR.

a) *Caption Generation*: Design a prompt template to guide the MLLM to obtain a brief caption for each image under constrained conditions, where type and  $k$  are two dataset-specific parameters to simulate the type and length of modified text in the real dataset.

b) *Image Pair Matching*: Match two image-text pairs to generate a quadruplet. Introducing a uni-modal image encoder to get the representation of every image and calculate the pairwise similarity between two different images. Then we can rank the similarities related to the reference image in descending order. Only one image whose similarity rank is between  $[c_0, c_1]$  ( $c_0 < c_1$ ) will be chosen as the target image, where  $c_0$  and  $c_1$  are two hyperparameters. This forms quadruplets.

c) *Modified Text Generation*: From the quadruplets, design a prompt template that generates modified text by either one of these:

- $P_{\text{temp}0} : \{C_t^i\}$  instead of  $\{C^i\}$
- $P_{\text{temp}1} : \text{Unlike } \{C^i\}, \text{ I want } \{C_t^i\}$
- $P_{\text{temp}2} : \{C_t^i\}$

d) *Positive Example Construction*: Combine image pairs from the second step with the modified text obtained in the third step to get new  $M$  triplets.

5) *Scaling Negative Examples*: Construct negative examples by replacing any element in the triplet. There are four

methods to construct negative examples, and the authors examined each method to conclude that replacing the target image will obtain the best negative examples.

6) *Two-stage Finetuning*: In the first stage, fine-tune both the query encoder and the target encoder with in-batch negative sampling. In the second stage, freeze the target encoder and only fine-tune the query encoder.

## B. Potential Approaches

We aim to improve the baseline model by integrating new techniques and optimizations.

1) *Image Generation*: One potential approach is to generate images that seamlessly align with the reference image and relative captions. By integrating a new image, we think that the model can incorporate image-side information to improve the performance of the model. We propose to input the reference image and text into the Multi-Instance Generation Controller (MIGC) [22] to generate a new image. This new image can provide useful information that are difficult to express in text at a fine-grained level.

2) *Two-stage Optimization*: We also propose a two-stage approach to optimize the performance. First, we filter out the candidate images that are not relevant to the reference image and modified text. Then, we use the filtered candidate images to retrieve the target image. This approach can reduce the search space and improve the retrieval performance.

## IV. EXPERIMENTS

We are currently conducting experiments to replicate the results of the original papers and to evaluate the impact of various learning techniques on retrieval accuracy. We intend to conduct additional experiments in the future following the implementation of certain improvements.

### A. Datasets

- **FashionIQ**: At present, we are utilizing only the *FashionIQ* dataset [12], as employed by the original study. FashionIQ is a natural language-based interactive fashion product retrieval dataset. It contains 77,684 images crawled from Amazon.com, covering three categories:



Fig. 2. Illustrative examples of the model's performance. The first two images represent success cases, while the last two images represent failure cases.

Dresses, Tops & Tees, and Shirts. Among the 46,609 training images, there are 18,000 image pairs. Each pair is accompanied by an average of two natural language sentences that describe one or multiple visual properties to modify in the reference image, such as “is shiny” or “is blue in color and floral, and with white base.”

- **Fashion200K:** We anticipate incorporating the *Fashion200K* dataset [23] in subsequent experiments. Fashion200K is a large-scale fashion dataset crawled from multiple online shopping websites. It contains more than 200,000 fashion images collected for attribute-based product retrieval, covering five categories: Dresses, Jackets, Pants, Skirts, and Tops. Each image is tagged with descriptive text corresponding to a product description, such as “beige v-neck bell-sleeve top.”
- **Shopping100K:** We also plan to use the *Shopping100K* dataset [24] in future experiments. Shopping100K is a large-scale fashion dataset of individual clothing items extracted from different e-commerce providers. It contains 101,021 images of 12 fashion attributes, covering categories such as “collar,” “color,” “fabric,” “fastening,” “fit,” “gender,” “length,” “neckline,” “pattern,” “pocket,” “sleeve length,” and “sport.” Compared to FashionIQ and Fashion200K, the Shopping100K dataset is more diverse and only contains garments in isolation.

### B. Evaluation Metrics

We employed *Recall@K* ( $R@K$ ) [25] as the primary evaluation metric, which measures the proportion of queries for which the retrieved top  $K$  images include the correct target image. *Recallsubset@K* ( $R_{subset}@K$ ) is a similar metric, but the model retrieves images only within the semantically similar group of the reference image.

### C. Current Results

We obtained results that are highly consistent with those reported in the original paper. To demonstrate the performance of our model, we present four images illustrating success and failure cases (see Figure 2).

## V. CONCLUSION

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